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FLORIDA TODAY

In Melbourne:
Florida Today is a 90,000 daily circulation newspaper on Florida's Space Coast. Visit its Web site at **www.floridatoday.com**

Executive editor: Derek Osenenko
Managing editor: Bob Stover
Project presentation editor: Tom Clifford
Project online editor: Traci Bauer
Design: Rob Landers
Reporters: Jennifer Ellis, Linda Jump, Chris Kridler, Wayne T. Price, Jeff Schweers, John Torres, Jim Waymer
Contributing editors: Steve Arnold, Steve Dozier, Alice Garwood, Maggie Erickson,Tom Kehoe, Chuck McClung, Joelle Moran, Matt Reed
Graphics: Octavio Diaz, Bobby Freeman, Jennifer Joslin
Photographers: Michael R. Brown, Rik Jesse, Craig Rubadoux
Online producers: Julian Beckford, Dave Larimer, Eric Naze, Lee Nessel
Researcher: Belinda Kehoe

THE NEWS-PRESS news-press.com

In Fort Myers:
The News-Press is a 90,000 daily circulation newspaper on the southwest coast of Florida. Visit its Web site at: **www.thenews-press.com**

Executive editor: Kate Marymont
Managing editor: Ronnie Ramos
Project editor: Cindy McCurry-Ross
Design and art direction: Jeremy Gilbert
Reporters: Betsy Clayton, Kevin Lollar, Dick Hogan
Contributing editors: Michelle Hudson, Eric Claro
Project photographer: Andrew West
Contributing photographers: Clint Krause, Marc Beaudin, Garth Francis
Photo editors: Gary Ulik, Mark Hume

News-Journal

In Pensacola:
The Pensacola News Journal is a 64,000 daily circulation newspaper in the Florida Panhandle. Visit its Web site at: **www.pensacolanewsjournal.com**

Executive editor: Randy Hammer
Managing editor: Bob Bryan
Contributing editor: Tom Ninestine
Reporter: Anton Caputo, Michael Suchcicki
Photographers: Michael Spooneybarger, Bruce Graner, Gary McCracken, Karena Cawthon

GANNETT NEWS SERVICE USA TODAY

Reporters: Alisa LaPolt and Paige St. John in Tallahassee, and Larry Wheeler in Washington

NEWS 10

In St. Petersburg-Tampa:
WTSP is Gannett's CBS affiliate covering the greater Tampa Bay area:. Visit its Web site at **www.wtsp.com**

News director: Lane Michaelsen
Reporter: Bill McGinty
Photojournalists: Tom Walters and Gene Yagle

FIRST COAST NEWS NBC 12

In Jacksonville:
WJXX and WTLV are Gannett's ABC and NBC affiliates covering Florida's First Coast. Visit their Web site at: **www.firstcoastnews.com**

News director: Mike McCormick
Meteorologist: Cliff Michaelsen

About this project: Joining forces for a closer look

Gannett newspapers and televisions stations, with the help of Gannett News Service bureaus in Tallahassee and Washington, have teamed up to produce "Paradise at Risk: Florida's Costly Battle with Beach Erosion."

The project brought together 67 journalists at three Florida newspapers, three TV stations and two news bureaus.

They produced an in-depth report about erosion and its impact on Florida lifestyles, politics, economics and environment.

Throughout the three-month process, reporters and photographers traveled more than 1,800 miles along Florida's coasts.

Members of the project team reviewed more than 2,000 state and federal documents, studied more than 90 Web sites and interviewed more than 100 sources, including people in each of Florida's 35 coastal counties.

The team consulted scientists, coastal engineers, professors, developers, residents, tourists, snowbirds, legislators, lobbyists and others. The collaboration has resulted in:

- This 16-page report, distributed in Gannett Florida newspapers in Fort Myers, Pensacola and Melbourne and in *USA Today* at locations in Tampa-St. Petersburg and Jacksonville.
- A dedicated "Paradise at Risk" Web site at **www.floridacapitalnews.com/erosion/index.html**
- A series of television reports, available to viewers in Tampa and Jacksonville and others through the Web site.



Tony Giberson, Pensacola News Journal

Encroaching ocean: Waves lay claim to a Pensacola Beach home, proof the erosion problem has become a crisis in many areas of the state.

On the Web

Log on to beach erosion at the "Paradise at Risk" site to find these Web extras:

- An interactive Florida map with county-by-county breakdowns of critical erosion rates, cost for renourishment projects, miles of sandy beach and the number of permits issued for buildings along the beach.
- An interactive quiz that tests your knowledge of different kinds of sand along Florida's beaches.
- An interactive graphic that describes how beach erosion happens.
- Several photo galleries that include current and historic snapshots submitted by readers throughout the state.
- Audio clips of key sources and staff involved in the project.

Log onto **www.floridacapitalnews.com/erosion/erosionindex.html**

Erosion impacts Florida from coast to coast



ESCAMBIA COUNTY
Pensacola Beach's first nourishment project begins this fall, when 4 million cubic yards of sand — $20 million worth — will be pumped from the Gulf of Mexico to add about 100 feet of frontage.

DUVAL COUNTY
Since 1975, six major beach renovations, costing $38 million, have taken place along Jacksonville beaches. Minor renourishment has maintained its current status.



BREVARD COUNTY
The county wants to renourish about seven miles in Central Brevard, including Satellite Beach, Indian Harbour Beach and unincorporated areas. But the federal government is trying to avoid disturbing marine life found in near-shore rocks.

GULF COUNTY
Home to St. Joseph Peninsula State Park, named one of the best beaches in America, but the peninsula area erodes so quickly that the state Department of Environmental Protection has deemed it impossible to renourish.



MANATEE COUNTY
About 12 miles, or 98 percent of the county's beach, are critical erosion areas. An $8.5 million project replenished sand on Anna Maria Island beaches this summer.

PINELLAS COUNTY
Among Florida counties, Pinellas ranks third in the amount of money spent on building up beaches:
$101.7 million.
The next renourishment project is scheduled for spring 2003. About $1.7 million will be spent to spread sand across 3,000 feet of Upham Beach in St. Pete Beach and one mile of Sunset Beach in the city of Treasure Island.

SARASOTA COUNTY
A $4.2 million renourishment project successfully rebuilt Lido Beach in Sarasota County, from 30 feet to 200 feet. But many residents complained that the sand was too dark and hot; it will be replaced this year.



LEE COUNTY
An $11.4 million project is planned for about four miles on Estero Island and 1.1 miles on Lovers Key. A delay in federal funds has put the project in limbo. However, an emergency $60,000 project, with sand from an inland quarry, was completed last month.



MIAMI-DADE COUNTY
More renourishment money has been spent in Miami-Dade than in any other Florida county. Beaches average 200 feet wide, with a stretch of 580-foot-wide sand at South Beach.

BROWARD COUNTY
Nearly all of Broward County's coastline is eroding, though some areas appear OK, and just 35 percent fall into the state's critical erosion category.
This fall, a $44.92 million project is expected to launch the widening of 12 miles of beach in Pompano Beach, Fort Lauderdale, Hollywood, Lauderdale-by-the-Sea and Hallandale Beach.

BY THE NUMBERS

0
Number of Florida beach renourishment projects requested of Congress by the White House.

$30 million
Amount Florida budgeted this year for beach nourishment projects and studies.

825
Miles of sandy beach in Florida.

162
The number of miles of Florida beaches that have been rebuilt.

$17.7 billion
Amount spent by Florida's beach-going tourists in one year.

4,913
The number of homes, hotels, condominiums and other buildings built on shores subject to erosion since 1978.

52
The number of permits denied since 1978.

$886 million
The amount of 2002 dollars spent on rebuilding Florida beaches in the past 79 years.

$132 million
Federal money spent rebuilding Florida beaches since 1995.

Sources: Florida county officials, Florida Department of Environmental Protection, Congressional records, Florida Legislative records, Florida Shore & Preservation Association



Andrew West, The News-Press

Pounding sea: The ocean pummels Rosanna Mata, left, Patricia Urquijo and her son, Carlitos, 3, as they walk north on the beach in May in front of the Westin Diplomat Resort and Spa in Hollywood in Broward County. "When we walk, it's dangerous. You have to watch your step," Urquijo said. Sand pumped in by the resort in January washed away within days.

State defends shrinking shore

As tides continue to devour the coast, development fuels renourishment projects

SHORE: Continued from Page 1

Gino Angella has seen the power of erosion.

The Miami native bought his dream home across the street from the beach in south Broward County two years ago. Since then, waves have pummeled the house across the street, causing it to crumble into the surf.

"This neighborhood has been here 50 years," he said. "I'll have to pray it'll be here another 50 years."

Why isn't Angella panicking?

Because like dozens of coastal communities around the state, his is slated for a beach renourishment project in the next year or two.

Nourishment — which usually involves dredging sand from the nearby ocean floor to rebuild a beach — has turned the state into a mecca for coastal engineers and dredging companies that profit from the projects.

With near-shore sand becoming scarce in some areas, particularly Southeast Florida, many expect renourishment to get more expensive.

Some see it as a fool's errand — fighting rising sea levels, hurricanes and other forces that can turn a mountain range to dust given enough time. Throw in Florida's habit of dredging inlets to support shipping and boating, which adds to the natural forces eating at the shore, and the battle can seem insurmountable.

> "We could quit getting the sand and watch the coast turn to concrete. Who knows, maybe we'll get to the point in Florida where there are only pockets of beach left."
>
> **STEPHEN LEATHERMAN,** "Dr. Beach," director, Florida International University Laboratory for Coastal Research

Others see it as the only way to protect the state's vast tourism industry and high-dollar coastal real estate. Consider: While beach renourishment has cost taxpayers $886 million since the early 1920s, the state's beaches generate $17.7 billion in direct tourist spending every year, according to the Florida Shore and Beach Preservation Association, which lobbies lawmakers for money for beach projects.

"If you look at the numbers, renourishment is the best investment you can make," said Bob Dean, graduate research professor of coastal and oceanographic engineering at the University of Florida.

Luis Garcia, 17, doesn't know much about beach renourishment, but his actions agree with Dean's assessment. The Palm Beach County resident will drive more than an hour to find a wider beach. He and his buddies head to Miami Beach, a 500-foot-wide sandy wonderland that is a renourishment success story.

"I like it here," Garcia said, tossing a football. "It's much bigger."

But to Walton County resident Celeste Cobena, renourishment spawns an expensive, never-ending cycle without offering a solution.

The cycle: Renourish the beach to protect development, and make the area more appealing to further development. Once that development comes — and it does — the need for renourishment increases.

"We need to go back and look at the root of the problem, and the root is building too close to the beach," said Cobena, who is worried about the environmental and aesthetic impact renourishment could have on the Panhandle's pristine beaches. "People build too close and then want the taxpayers to bail them out."

'It's probably too late'

The state instituted a setback line in 1978 to limit development near the ocean, and strengthened those controls in 1985 when it began phasing in the Coastal Construction Control Line.

But the law contains enough exemptions that thousands homes, hotels and condos have been built on Florida shores, seaward of the line. Without renourishment, many could wash away.

Orrin Pilkey, a Duke University geologist who specializes in the study of developed shorelines, believes Florida's is one of the worst beach management systems in the country. He'd like to see coastal communities retreat from the shore, and recommends the state and the federal government, which has spent $3.6 billion on renourishment projects nationwide since 1923, instead use the money to buy back beachfront property.

"It's probably too late in Florida in many ways because you have a shoreline full of high-rises," he said.

According to Duke University, there were four renourishment projects in the 1950s. The number jumped to 36 in the 1960s, 59 in the 1970s and 85 in the 1980s. It dipped to 69 in the 1990s, but those projects accounted for a greater volume of sand than the projects of any preceding decade. The trend isn't likely to reverse soon. This fiscal year, which began July 1, 27 projects will get a portion of the $30 million the state sets aside for renourishment each year.

Those appropriations range from $50,000 for a feasibility study in north Volusia County to nearly $10 million to rebuild storm-ravaged Pensacola Beach.

Of the 27 projects, eight are in the construction, or beach building, phase. Last year, 26 received money, with six beaches rebuilt.

Questions raised

Pilkey worries escalating costs will make renourishment impossible in the next 50 years and wonders about environmental impacts of dredging tons of sand from the ocean floor.

Pilkey is part of an organized group of 70 scientists who worry about unknown, cumulative impacts and point to lessons learned from the decades-old practices of altering the Mississippi River's flood plain and filling the Florida Everglades. Both were historically accepted management methods that caused major environmental harm, and likely will cost taxpayers billions to fix.

"The bottom line is that we don't know what the impacts are in either the short or long term, and this is significant because there are a wide array of documents that conclude there are no impacts," said Ken Linderman, a senior scientist with Environmental Defense, an environmental advocacy group.

Dean, like many coastal engineers, disagrees, arguing that every study shows there are limited and temporary impacts.

He also thinks the doom and gloom forecasts for the future of renourishment are overblown.

To the contrary, he thinks Florida will see the number and size of projects decrease in the next 15 years as the state catches up with decades of erosion. He sees the explosion of renourishment projects in the past few decades as making up for years of neglect and thinks the state will eventually reach a sort of equilibrium with Mother Nature.

Help can't come too soon for South Florida, which is almost out of nearshore sand it depends on for beach building.

Without renourishment, many of the beaches that tourists flock to would simply not exist.

It all adds up to a high-stakes balancing act to Stephen Leatherman, director of Florida International University's Laboratory for Coastal Research, who is also known as "Dr. Beach," and is famous for his annual ranking of the world's and country's best beaches.

His big question: What's the alternative?

"We could quit getting the sand and watch the coast turn to concrete," he said. "Who knows, maybe we'll get to the point in Florida where there are only pockets of beach left."

Anton Caputo is a reporter for the Pensacola News Journal. *E-mail him at antoncaputo@pensacolanewsjournal.com*

Betsy Clayton, a reporter for The News-Press *in Fort Myers, and Paige St. John, a reporter with Gannett News Service in Tallahassee, contributed to this story.*



Andrew West, The News-Press

Dune dropoffs: Visitors to south Fort Myers Beach near Big Carlos Pass prepare to cross a small inlet. The beach is part of Lee County's 19.9 miles on the state Department of Environmental Protection's critical erosion list.

ON THE COVER

High tide with no beach

Like most mothers, Cristina Castro watches her 3-year-old every minute the girl plays in the surf in front of their Broward County apartment.

But Castro also has other things to watch: The sand buckets, shovels, beach umbrella and towels threaten to leave every time a wave crashes on the beach. The little girl, Korinna, chases into the water to retrieve her beach shovel every five minutes.

That's because the beach they visit isn't really a beach anymore.

The waves near the Ocean Crest apartment building crash on the shore and muscle up against the seawall — a structure that, ironically, was put there to prevent erosion.

But the beach is completely eroded.

When the Castros arrive at the Atlantic, they leave their flip-flops on the seawall's ledge. Leave them on the sand, and they'd be gone in minutes. They plunge the pole of the beach umbrella into the sand at the base of the seawall, understanding they'll have to reposition it several times before the afternoon fades because the waves lick at it and take sand from beneath it.

The mother and her child must leave before high tide.

At high tide, there is no beach.

The section of Broward County where they play is slated to be renourished — if federal dollars come through this year.

If not, they'll be forced to keep coming at low tide. They'll continue to fold their beach towels because there is not enough sand to stretch one out full length.

"I've never seen the beach like this now," said Castro, who gets to the beach via wooden stairs that descend from the seawall where a sign is posted. It reads, "Use stairs at your own risk. High tide covers the stairway."

"Before, it had a lot of space," Castro said. She moved to South Florida about 18 months ago. "You used to be able to spread a beach towel."

She looked around at the narrow strip of land and picked up a soaked towel.

"People love the beach, even if it's small," she said.

— Betsy Clayton, The News-Press